ALP-ttbar Toy MC v1.8: All-Pixel 3D Vertex Fit with Reduced Material — Preregistration and Results

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Toy study motivated by Alimena et al., arXiv:2610.38932, which extends Rygaard et al., JHEP 10 (2023) 138

What this document contains. Page 1: this overview. Pages 2–4: v1.8 post-registration results (doi:10.5281/zenodo.24034481). Pages 5–8: v1.8 preregistration, frozen on GitHub and Zenodo on 7 November 2026 before any v1.8 simulation or fit (doi:10.5281/zenodo.24028384). Both reports are reproduced unchanged.
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1. Why v1.8

For the lightest ALPs (17 MeV) the displaced e^+e^- pairs look almost like photon conversions in the tracker material, and the conversion veto costs a large factor in the expected limit (127× at $c\tau$ = 1 mm, 3.8 T). Two earlier preregistered results pointed the same way: in v1.6 only reducing the material helped at 17 MeV (127× → 65× → 35× for material scale f = 1, 0.5, 0.25); in v1.7 measuring z at every layer helped from 50 MeV on but gained only 7.8% at 17 MeV.

2. The preregistered plan (frozen 7 November 2026)

• **Grid:** 3 material scale factors (f = 1, 0.5, 0.25) × 2 z geometries (pixel layers only; all 14 layers) × 2 r–z covariances, on the unchanged v1.6/v1.7 samples and fits.
• **V-oracle:** a diagnostic covariance using the true electron momentum at every material surface, to test whether the v1.7 limit came from bremsstrahlung-reduced momenta (truth-level bound, not a realistic method).
• **Safeguards:** sanity gate (identity with v1.6 and v1.7), three independent validations (W1–W3), seven predictions (P1–P7) written before running.

3. Results (computed after the preregistration was public)

selection	17 MeV	50 MeV	100 MeV	200 MeV
transverse only, f = 1 (v1.6 baseline)	127×	68×	28×	8.2×
3D, z at every layer, f = 1 (= v1.7)	117×	38.5×	13.1×	3.9×
3D, z at every layer, f = 0.5	56×	17.9×	5.7×	2.2×
transverse only, f = 0.25 (v1.6)	35×	19.5×	8.8×	2.1×
3D, z at every layer, f = 0.25	**26.7×**	**8.8×**	**3.3×**	**1.1×**

• **Material and z coverage reinforce each other:** the 17 MeV gain from z at every layer grows from 7.8% to 14% to 23% as the material is reduced.
• **V-oracle** removes most conversion fakes (7.4% → 1.9%) but improves 17 MeV only from 117× to 113×: the 17 MeV limit is set mainly by multiple scattering relative to the 0.42 mrad opening, which revises the v1.7 interpretation.
• **Predictions:** P1–P5 and P7 confirmed; P6 (17 MeV ≤ 105× with V-oracle) refuted. Gate and W1–W3 passed; no deviation in the method.

4. Next steps (each under a public preregistration)

Layer geometry (a first layer closer to the beam, more and thinner layers, as in next-generation concepts such as ALICE 3 and ITS3; no performance claim); a joint 3D vertex fit; a dimuon control study; a Gaussian-sum filter, now lower priority.

Records

record	DOI / repository
v1.8 post-registration (pages 2–4)	https://doi.org/10.5281/zenodo.24034481
v1.8 preregistration (pages 5–8)	https://doi.org/10.5281/zenodo.24028384
code	github.com/wspaik5/ALP_ttbar_Toy_MC_v1.8_Postregistration_AllPixel_Reduced_Material